

COURSE OVERVIEW

1. **OPC UA Architecture:** Every major automation vendor sat on the committee that designed OPC UA, which is why it took twelve years and why it works with all of them. The unified architecture replaced DCOM — a Windows-specific middleware that died when Windows security policies tightened — with a transport-independent, platform-neutral stack. Client/server and pub/sub coexist because different applications need different patterns.
2. **Information Model:** An OPC UA server exposes a graph of nodes, each with a NodeID, a type, and references to related nodes. A client can browse this graph on any server and understand what data is available without vendor documentation. Address space structure, node classes, and the browsing sequence that makes OPC UA self-describing.
3. **Services:** Read, Write, Browse, Subscribe, and CreateMonitoredItems replace the custom driver development that OPC Classic required for every new vendor. The full service set, and which ones appear in every integration versus which ones rarely matter in practice.
4. **Sessions and Channels:** SecureChannel handles the encrypted transport. Session carries user authentication and context on top of it. Channel negotiation, session establishment, token lifetime, and what happens to active subscriptions when a channel drops and reconnects.
5. **Security:** Most production OPC UA deployments run Security Mode None because commissioning happened under a deadline and no one changed it afterward. Mode Sign, Mode SignAndEncrypt, certificate exchange, PKI structure in a plant environment, and why leaving security off is a larger problem than it appears.
6. **Subscriptions:** A client creates a subscription with a publishing interval, then adds MonitoredItems with individual sampling intervals and deadband filters. The full configuration chain that determines how much network traffic an OPC UA connection generates and how much data lands in the historian.
7. **Data Types:** Scalar types, structures, enumerations, arrays, and the QualityStatusCode that tells a client whether a value is valid, stale, uncertain, or from a failed sensor. The type system is what allows OPC UA to represent any process value without custom encoding.
8. **Discovery:** Local Discovery Server and Global Discovery Server allow clients to find OPC UA endpoints without being given addresses in advance. Endpoint enumeration, certificate trust during discovery, and the security gap that comes with an open discovery infrastructure.
9. **Implementation:** Choosing between open62541, the Unified Automation SDK, and platform SDKs for Siemens and Rockwell. Configuring Ignition as an OPC UA server. Common integration failures and how to diagnose them with the OPC UA error model and Wireshark together.
10. **Protocol Lab:** A Wireshark capture of a full OPC UA session from TCP connection through SecureChannel negotiation, browse, subscription, and disconnect. Reading the capture to verify security mode, subscription timing, and the exact moment a session token expires and renews.

Certification Relevance

- **OPC Foundation Certified** — Developer and Integrator exams test the information model, service set, and security configuration directly.
- **GICSP (GIAC)** — OPC UA security modes and integration patterns appear in ICS security exam prep domains.
- **ISA CCST** — Industrial networking and system integration standards tested at all CCST levels.

Next Steps

1. Pair OPC UA with the Security guide for the strongest overlap with GICSP exam domains.
2. Download open62541 (free, open-source C stack) and run a local server against UA Expert client.
3. OPC Foundation spec Parts 1 (Concepts) and 4 (Services) are the normative reference.
4. Apply at opcfoundation.org for the OPC UA Integrator certification exam.